

JATOTH SHYAMKUMAR

EMBEDDED SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of experience in C and Embedded C, simulator development, GUI applications, software debugging, functional testing, data analysis, and technical documentation. Experienced in developing Mission Computer and Bus Monitor simulators using Qt Creator and Visual Studio. Hands-on experience with MIL-STD-1553B communication, DDC device drivers, BC-RT configurations, real-time data monitoring, timers, and Hardware-in-the-Loop (HIL) simulation. Experienced in code traceability, software validation, and preparation of ICD, SRS, STR, STP, and VDD documentation.

TECHNICAL SKILLS

Programming:	C, Embedded C
GUI & Development:	Qt Creator, Visual Studio
Communication:	MIL-STD-1553B, RS-422 Serial Communication
Hardware & Drivers:	DDC Device Drivers
Testing:	Functional Testing, Debugging, Code Traceability, Code Validation
Data & Analysis:	MATLAB, Data Analysis
Simulation:	Mission Computer Simulator, Bus Monitor Simulator, Hardware-in-the-Loop Simulation
Other:	Timers, Scheduling, DIP/DOPs, Technical Documentation

PROFESSIONAL EXPERIENCE

RESEARCH CENTRE IMARAT (RCI), HYDERABAD

Contract Software Engineer | February 2022 – Present

- Develop and maintain simulator applications for testing On-Board Computer (OBC) software.
- Develop GUI-based simulator applications using Qt Creator and Visual Studio.
- Implement real-time parameter display and monitoring capabilities within simulator applications.
- Work with DDC drivers for receiving and transmitting MIL-STD-1553B data.
- Implement and work with Bus Controller–Remote Terminal (BC-RT) configurations for simulator communication with OBC.
- Implement timer-based functions for executing simulator operations at defined time intervals.
- Perform data analysis and monitoring of MIL-STD-1553B communication data.
- Perform software debugging, functional testing, code traceability, and validation activities.
- Participate in RIG Test, Carriage Flight Trial, and Release Flight Trial activities.
- Contribute to software code validation activities conducted in front of RCMA and R&QA; teams.
- Prepare and manage technical documentation including ICD, SRS, STR, STP, and VDD.

KEY PROJECTS

GUI FOR MISSION COMPUTER SIMULATOR

Tools/Technologies: Qt Creator, DDC Drivers, MIL-STD-1553B, Timers, Hardware-in-the-Loop Simulation

Project: Mission Computer Simulator for testing OBC software.

- Collected and analyzed basic simulator requirements and developed the GUI using Qt Creator.
- Implemented real-time parameter display and monitoring.
- Integrated DDC drivers for data transmission and reception through the hardware interface.
- Implemented BC-RT communication configuration with the simulator acting as Bus Controller and OBC as Remote Terminal.
- Implemented timers for executing simulator functions at different time intervals.
- Implemented a six-point Launch Acceptance Region display for monitoring the real-time weapon-launch scenario.
- Supported Hardware-in-the-Loop Simulation.

GUI FOR BUS MONITORS

Tools/Technologies: Visual Studio, DDC Drivers, MIL-STD-1553B

Project: Bus Monitor simulator for monitoring MIL-STD-1553B data from different buses and displaying parameters in real time.

- Collected and analyzed basic requirements for Bus Monitor applications.
- Developed Telemetry Bus, EW Bus, and Avionics Bus Monitor simulators using Visual Studio.
- Integrated DDC drivers for receiving communication data.
- Implemented the simulator as a monitoring terminal with GUI-based data display.
- Supported analysis and monitoring of MIL-STD-1553B communication data.

EDUCATION

M.Tech – Embedded Systems | 2014–2018 | 71.81%

Swami Vivekananda Institute of Technology, Hyderabad – JNTUH

B.Tech – Electronics & Communication Engineering | 2011–2014 | 64.90%

Keshav Memorial Institute of Technology, Hyderabad – JNTUH

Diploma – Electronics & Communication Engineering | 2008–2011 | 69.63%

Sana Polytechnic College, Kodad – SBTET

SSC | 2007–2008 | 62.83%

Sharadha High School, Thorrur